

RYE HOWARD-STONE

github.com/rhowardstone — [LinkedIn](#) — [Google Scholar](#)

Computational scientist with a PhD in Computer Science and dual undergraduate degrees in Mathematics and Physics. I apply statistical and data science methods to large, noisy datasets and build the tooling that turns them into actionable insight. Experience instrumenting and optimizing AI-assisted developer workflows.

Selected Projects

CaseStack Consulting — Independent Jan 2026 – Present

- Built a full-text analytical platform capable of processing millions of pages of government records using Python, SQLite/FTS5, and Datasette. Results used by reporters at major outlets, and high-impact independent outlets. The website hosting the proof-of-concept corpus results now gets ~15K DAU.
- Designed quantitative methodologies for the test corpus: Bates-stamp serial-number gap analysis, redaction-pattern statistics, and alteration categorization across ~47K modified documents. Published 190+ analytical reports.
- One-person data/AI team: utilized Claude Code for ingestion, OCR/transcription correction, cross-referencing, targeted PII redaction across ~2,400 pages, and forensic audit of ~3M documents. Created integrated AI chatbot using free models from OpenRouter, reverse image search using CLIP embeddings, MCP server, and much more.

AI Agent Systems for Research Workflows — Independent 2025

- Built [claude-code-scientist](#), a semi-autonomous AI research agent with 24 specialized skills, validation hooks, and provenance tracking across multiple literature APIs.

AmpliconHunter / AmpliconHunter2 — UConn 2024 – 2025

- Created an *in silico* PCR tool scalable to multimillion-genome databases (ICCABS 2025). Rewrote core engine in C with AVX2 SIMD parallelization for 5.9x throughput improvement (ICCABS 2026).

Open-Source Performance Contributions 2024 – 2025

- Submitted performance optimization PRs to MDAnalysis, scikit-bio, and Biopython. [Cython rewrite](#) for MDAnalysis (10 to 18x speedup) under review.

Experience

Software Engineer — Early-Stage Consumer Tech Startup 2025 – Present

- Contracted to contribute to the technical development of a pre-launch web platform under NDA. Work included prototype implementation, AI integration, and collaboration with the founder and a second developer on product architecture and interaction design.

Graduate Research & Teaching Assistant — University of Connecticut 2019 – 2025

- Led statistical analysis of the largest long-read PacBio CCS microbiome dataset of its kind at the time, in collaboration with microbiologists, veterinarians, and other scientists at Pfizer and Intus Biosciences. Designed the analysis plan, ran the modeling, and translated findings for a multidisciplinary audience. Published in *Frontiers in Microbiology*.
- Primary instructor for Ethics in Computer Science (CSE 3000), including coding projects on quantifying bias in algorithms. TA for Data Structures, Systems Programming, Bioinformatics, Computational Genomics, and Intro to Software Engineering.

Education

PhD, Computer Science & Engineering — University of Connecticut December 2025

Dissertation: Computational Methods for High-Resolution Microbiome Profiling. Advisor: Ion Mandoiu. Coursework: machine learning, AI for software development, data mining, algorithms, statistical methods.

BS Mathematics (Applied) & BS Physics (Engineering) — Southern Connecticut State University May 2017
Computer Science minor. Pi Mu Epsilon (National Mathematics Honors Society).

Technical Skills

Languages: Python, R, SQL, C/C++, Matlab/Octave, JavaScript, Java

Statistics & ML: experimental design, regression, clustering, dimensionality reduction, NLP, sequencing analysis

Data & Tooling: SQLite/FTS5, Datasette, Pandas/NumPy, Linux, Git, HPC, Docker, AI agent frameworks

Selected Publications

Howard-Stone, R., *et al.* (2025). Cecal microbiome transplantation without antibiotic preconditioning standardizes murine microbiomes. *Frontiers in Microbiology*.

Howard-Stone, R. & Măndoiu, I.I. (2025/2026). [AmpliconHunter](#) & [AmpliconHunter2](#). *ICCABS post-proceedings*.

Howard-Stone, R. & Măndoiu, I.I. (2025). MHASS: Microbiome HiFi Amplicon Sequencing Simulator. *Bioinformatics*.